

ECE20007 Final Project: Audio Equalizer

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Design and Implementation of an Analog Audio Equalizer

1.2 Definition and Purpose of the Project

- The primary objective of this project is to transition analog signal processing theory into a functional experimental design.
- Specifically, the project utilizes filters and operational amplifiers to construct an audio equalizer capable of modifying an analog audio source.
- The system is designed to take a standard, non-amplified audio signal, and allow the user to independently boost or reduce low, middle, and high-frequency bands.
- Ultimately, the processed signal is fed into a power amplifier capable of outputting at least 400mW to successfully drive an 8 ohm speaker system.

1.3 Key Functions of the Audio Equalizer

- **Input Stage:** Distributes the initial analog audio signal into parallel filtering paths.
- **Filtering Stage:** Utilizes specific filter topologies to accept and isolate distinct low, mid, and high-frequency bands.
- **Adjustment Stage (Volume Control):** Employs isolated potentiometers to selectively reduce the amplitude of the outgoing filtered waveforms without loading the circuits.
- **Recombination Stage:** Uses a summing amplifier to add the adjusted frequency signals back into a single combined output.
- **Output Stage:** Utilizes a class AB power amplifier to increase the current capacity of the signal, providing the necessary wattage to physically drive a speaker cone.

1.4 Applications of Audio Equalizers

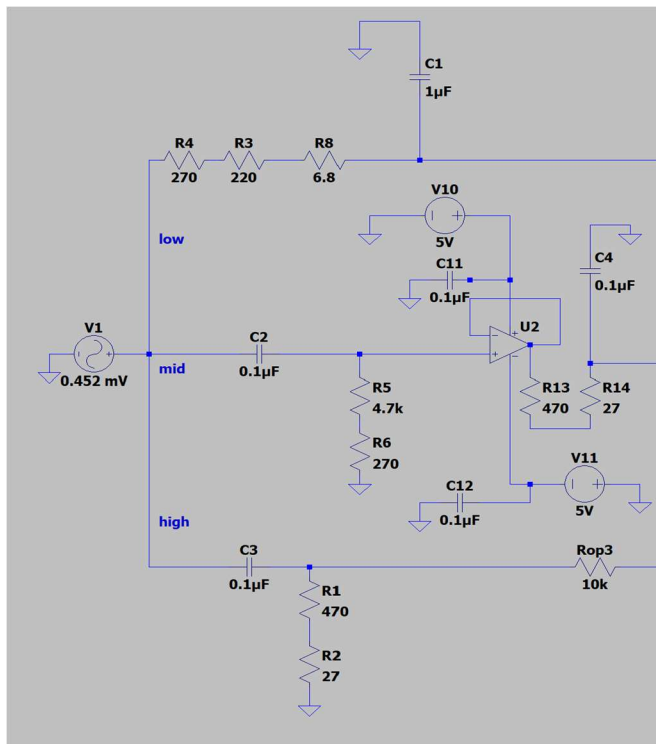
- Audio equalizers are ubiquitous in consumer and professional electronics used to tune frequency responses to user preferences or environmental acoustics.
- Common everyday applications include mobile phones, Bluetooth speakers, car radios, and musical instrument amplifiers.

2. Theory

2.1 General Filters Working Principle

Passive filters operate entirely on circuit reactivity, meaning the components respond to frequency changes by returning power to the source rather than utilizing an external powered element.

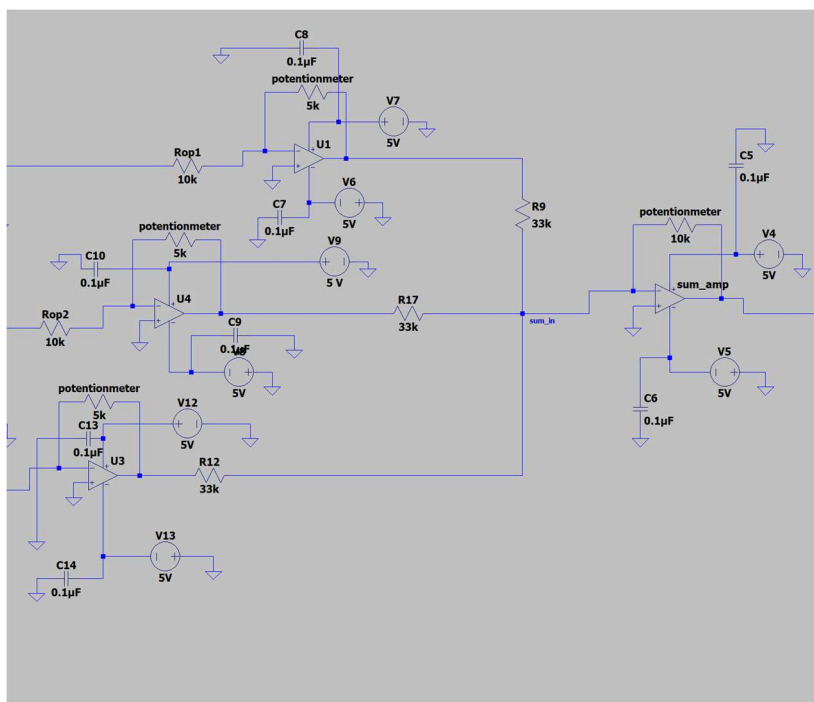
- **Low-Pass Filter:** Permits frequencies below a designated cutoff frequency (f_c) to pass while attenuating higher frequencies.
- **High-Pass Filter:** Permits frequencies above a designated cutoff frequency (f_c) to pass while attenuating lower frequencies.
- **Mid-Pass Filter:** Functions as a combination of high-pass and low-pass characteristics to isolate a specific intermediate frequency band.
- **Cutoff Frequency Equation:** For standard first-order RC filters, the -3dB cutoff frequency is determined by the formula: $f_c = 1/(2\pi RC)$



2.2 Operational Amplifier's Working Principle

Active filtering introduces operational amplifiers to filter topologies, which provides controllable gain, improved output isolation, and higher-order cutoff rates.

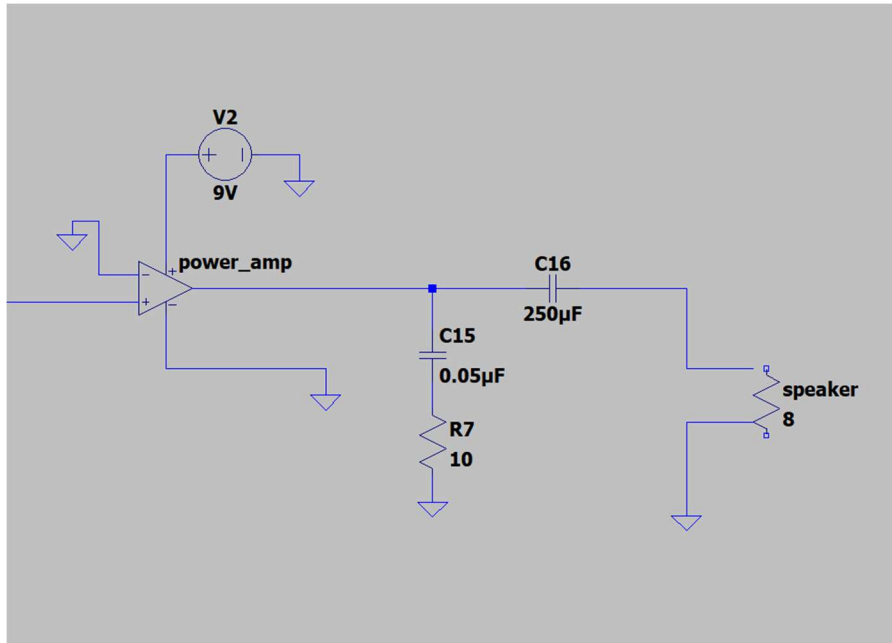
- **Buffering/Isolation:** Op-amps can be configured as voltage-followers (buffers) to isolate the input signal from the output. The buffer electrically recreates the input signal on the output terminal, dividing the system into separate circuits to prevent subsequent stages from loading the filters and altering their cutoff frequencies.
- **Summation Amplifier:** An inverting amplifier topology with multiple inputs connected to the negative terminal. Because op-amps utilize negative feedback to match their input terminals, grounding the positive terminal creates a "virtual ground" at the negative terminal. This prevents alternate conduction paths, allowing the amplifier to add multiple input signals together while largely ignoring superposition.
- **Summation Equation:** $V_{out} = -R_f(V_1/R_{in1} + V_2/R_{in2} + V_3/R_{in3})$



2.3 Power Amplifier's Working Principle

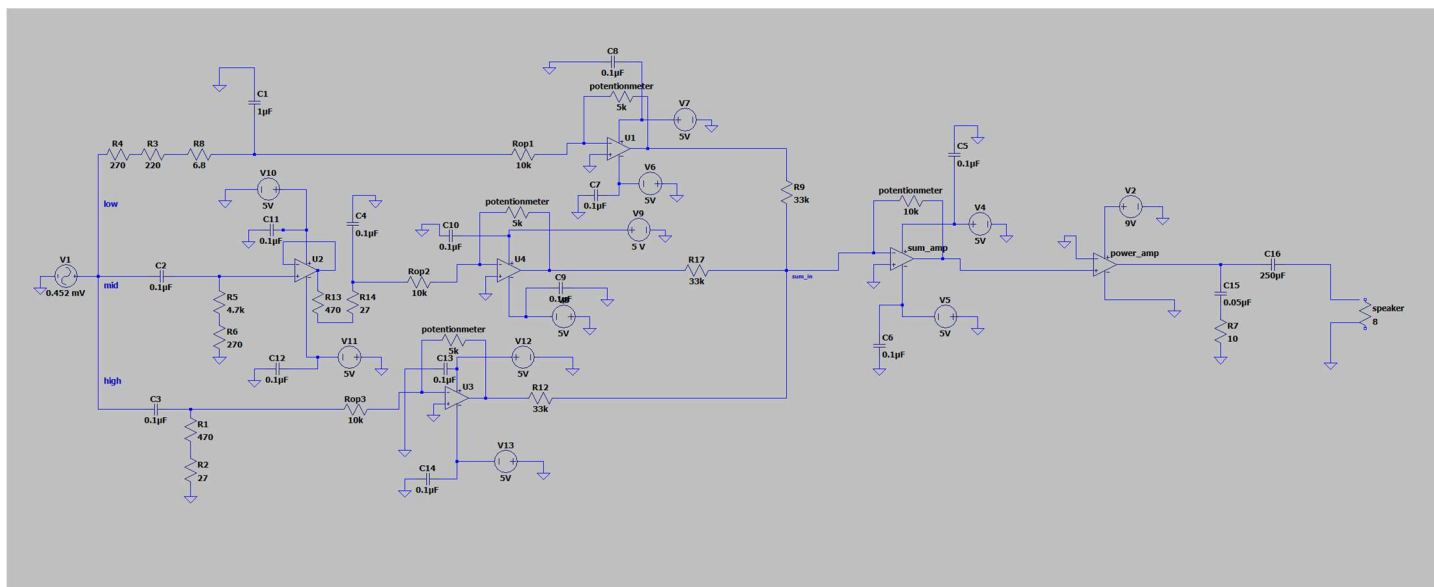
While standard operational amplifiers (like the LM324) and power amplifiers (like the Class AB LM386) share similar architectures, they are distinguished by their output current capacity. A standard op-amp may only output roughly 5.9 mA due to high output terminal

resistance, whereas a power amplifier can output upwards of 223 mA under the same load. This massive increase in current delivery is strictly necessary to generate the physical power required to drive a speaker cone.



3. Procedure

3.1 Schematic of the Audio Equalizer



3.2 Design Parameters

Note: Below are the theoretical calculations required to hit the project specifications. Ensure you update the standard component values (R and C) to match the exact physical components you used on your breadboard.

3.2.1 Filter design parameters

- **Low-Pass Filter:** Target -3dB cutoff is 320Hz +/- 10%
 - Choosing a standard capacitor $C_1 = 1 \text{ uF}$
 - $R_1 = \frac{1}{2\pi(320)(1.0 \times 10^{-6})} = 497.35 \text{ } \Omega$
 - I used the closest possible, $470 \text{ } \Omega + 27 \text{ } \Omega$ (in series) = $497 \text{ } \Omega$
- **High-Pass Filter:** Target -3dB cutoff is 3200 Hz +/- 10%
 - Choosing a standard capacitor $C_2 = 0.1 \text{ uF}$
 - $R_2 = \frac{1}{2\pi(3200)(1.0 \times 10^{-7})} = 497.35 \text{ } \Omega$
 - Ran out of $27 \text{ } \Omega$ resistors, used $270 \text{ } \Omega + 220 \text{ } \Omega + 6.8 \text{ } \Omega$ (in series) = $496.8 \text{ } \Omega$
- **Mid-Pass Filter:** Target range is 320 - 3200 Hz +/- 10%
 - Choosing standard capacitors (one for high pass, one for low pass part):
 - I used an op-amp as a voltage buffer to prevent input loading (LF356)
 - $C_1 = 0.1 \text{ uF}$
 - $C_2 = 0.1 \text{ uF}$
 - $R_1 = \frac{1}{2\pi(320)(1.0 \times 10^{-7})} = 4.97 \text{ k}\Omega$
 - $R_2 = \frac{1}{2\pi(3200)(1.0 \times 10^{-7})} = 497 \text{ } \Omega$

3.2.2 Volume control circuits design parameters

In this design, the filtered signal passes through a potentiometer configured as an adjustable voltage divider, which is immediately followed by an op-amp configured as a voltage follower (buffer).

- **Potentiometer Value:** $R_{\text{potentiometer}} = 10 \text{ k}\Omega$

- **Operation:** The potentiometer wiper sweeps from ground $0\ \Omega$ to the maximum potentiometer resistance value, outputting a fraction of the input voltage:
- $$V_{out} = V_{in} * \left(\frac{R_{wiper}}{R_{potentiometer}} \right)$$
- The subsequent op-amp buffer physically isolates this voltage divider from the summation amplifier, ensuring the final stage does not load the potentiometer and alter the intended volume level.

3.2.3 Signal recombination stage design parameters

- The system requires a maximum output of $100\text{ mV}_{\text{rms}}$ +/- 10% when driven by a $320\text{ mV}_{\text{rms}}$ input with all volume knobs maximized.
- Therefore, the recombination stage must apply an overall attenuation.
- The recombination stage was designed using an inverting summing amplifier. The three filtered and volume-adjusted signals were connected into the summing amplifier through input resistors. The schematic used: $R_1 = 33\text{ k}\Omega$, $R_2 = 33\text{ k}\Omega$, $R_3 = 33\text{ k}\Omega$. The feedback resistor used: $R_f = 10\text{ k}\Omega$. The summing amplifier equation is:

$$V_o = -R_f \left(\frac{V_{low}}{R_1} + \frac{V_{mid}}{R_2} + \frac{V_{high}}{R_3} \right)$$

- $$V_o = -10k \left(\frac{V_{low}}{33k} + \frac{V_{mid}}{33k} + \frac{V_{high}}{33k} \right)$$
- $$V_o = -0.303 (V_{low} + V_{mid} + V_{high})$$

This means the summing amplifier combines the three frequency bands while applying an overall gain of approximately 0.303 to each input path.

3.3 Short Description of Operation

The 320 mV input audio signal is split and routed into three parallel filter stages. These filters (low, mid, and high-pass) isolate their respective frequency bands. Because standard first-order filters are susceptible to input loading, operational amplifiers configured as buffers are used to isolate the filters from subsequent stages. The filtered signals then pass through op-amp-isolated potentiometers, acting as independent volume controls. These three signals are fed into a summation amplifier, which recombines them into a single waveform while setting the maximum voltage amplitude to the required 100

mV. Finally, this recombined signal is fed into an LM386 power amplifier, boosting the current output to drive the final 8 Ω speaker load with at least 400 mW of power.

4. Result

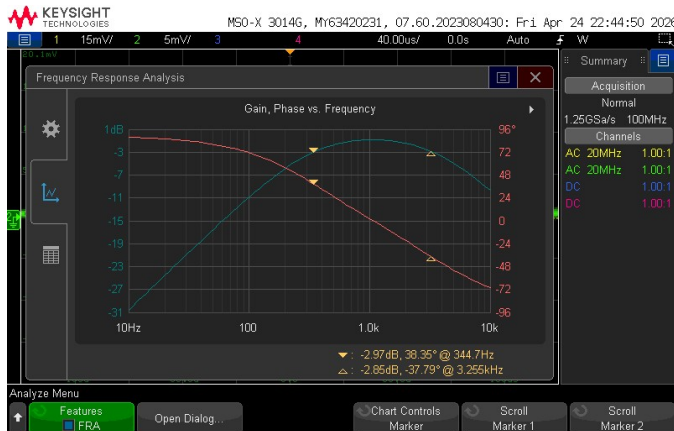
Instructions: Populate the subsections below with your oscilloscope screenshots, Bode plots, and bench measurements.

4.1 Low-pass filter's FRA curves with explanation



The FRA plot confirms the successful implementation of the low-pass bass filter. Low frequencies are passed with minimal attenuation (near 0 dB), while higher frequencies are gradually attenuated. The on-screen cursors indicate that the exact -3 dB cutoff point occurs between 316.2 Hz and 335.0 Hz, with a corresponding phase shift of approximately -45°. This measurement sits squarely within the required 320 +/- 10% tolerance range, successfully verifying that the filter correctly isolates the targeted bass frequencies

4.2 Mid-pass filter's FRA curves with explanation



The FRA plot confirms the successful implementation of the mid-pass filter, displaying clear band-pass behavior where both low and high frequencies are properly attenuated. The on-screen cursors indicate the exact -3 dB cutoff points occur at 344.7 Hz (lower) and 3.255 kHz (upper), accompanied by corresponding phase shifts of approximately +38° and -38°. Both measurements fall comfortably within the required +/- tolerance ranges for their respective 320 Hz and 3.2 kHz targets, verifying that the filter correctly isolates the mid-frequency band."

4.3 High pass filter's FRA curves with explanation

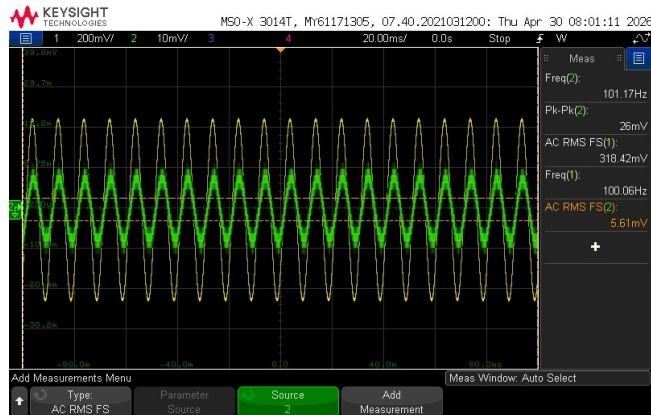


The FRA plot confirms the successful implementation of the high-pass filter. Low frequencies are strongly attenuated, while the gain steadily increases until it reaches the passband at higher frequencies (near 0 dB). The on-screen cursors indicate the exact -3 dB cutoff point occurs at 3.255 kHz, accompanied by the characteristic first-order phase shift of approximately +47° (46.79°). This measurement falls squarely within the required 3.255

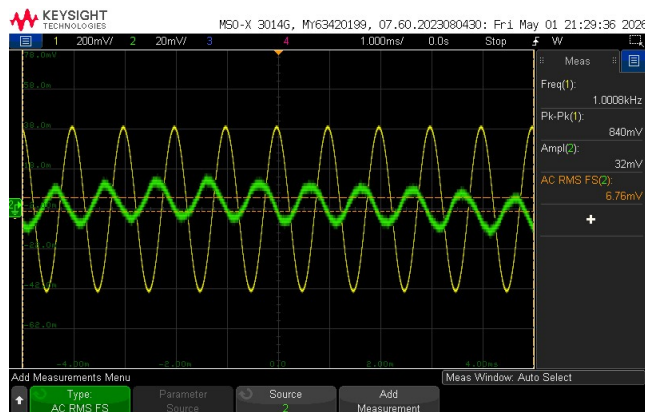
kHz +/- 10% tolerance range, successfully verifying that the filter properly extracts the treble frequency components.

4.4 Plots showing V_{amp} with all equalizers turned to minimum setting

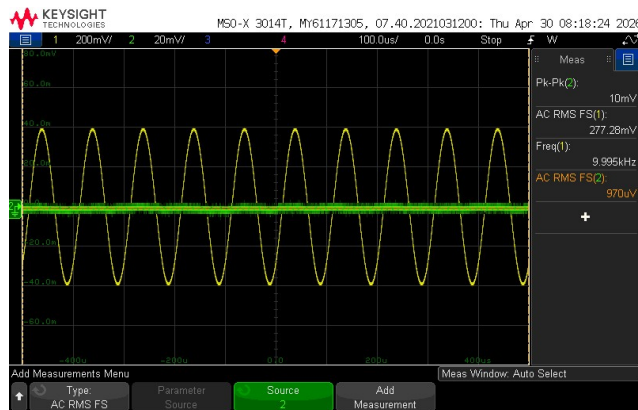
100 Hz



1k Hz



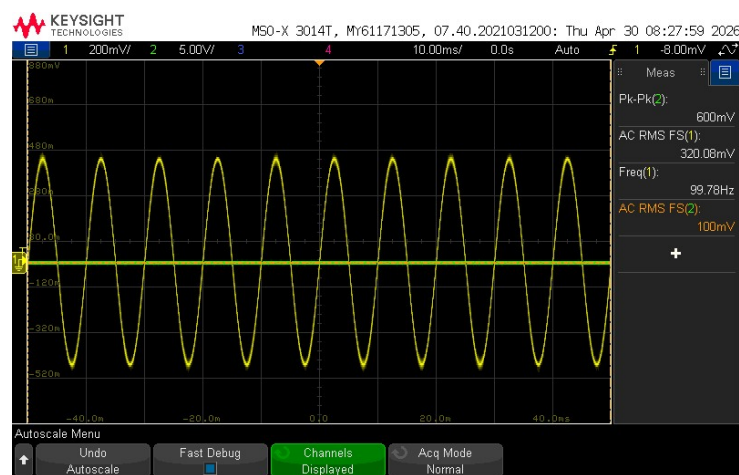
10k Hz



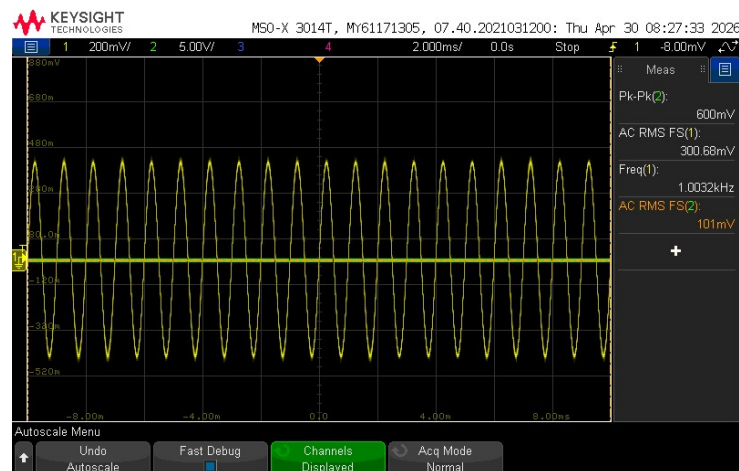
The oscilloscope plots at 100 Hz, 1 kHz, and 10 kHz with all equalizer knobs set to minimum show that the output voltage is significantly attenuated. At 10 kHz, the measured RMS voltage is approximately 1 mVrms, which is well below the required limit of 15 mVrms. Similar behavior is observed at 1 kHz and 100 Hz, where the signal is under the required limit of 15 mVrms. This demonstrates that the equalizer effectively reduces the output signal when set to minimum.

4.5 Plots showing Vamp with all equalizers turned to maximum setting

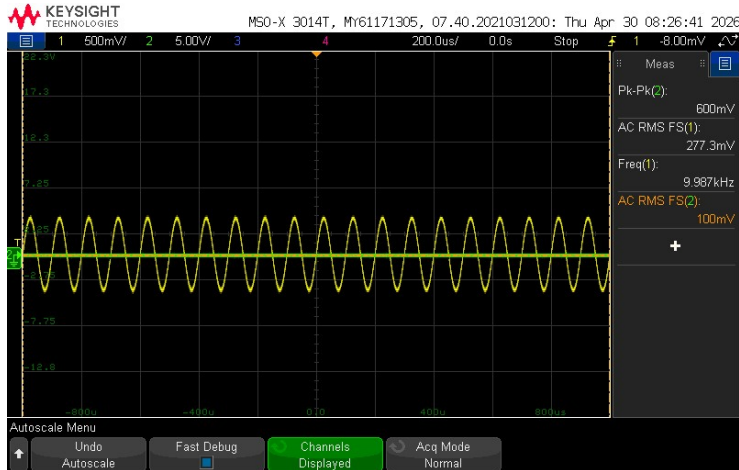
100 Hz



1k Hz



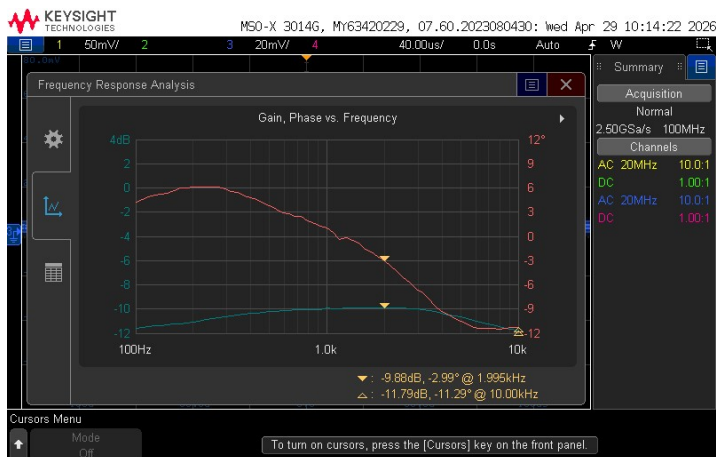
10k Hz



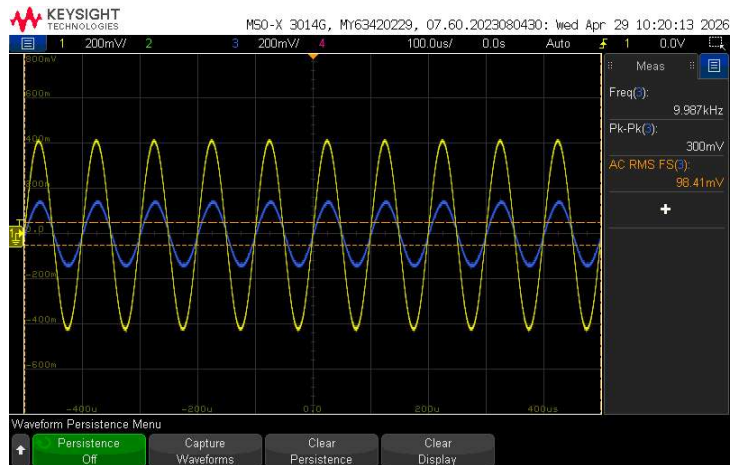
When all equalizer volume knobs are set to their maximum positions, the recombination stage successfully scales the voltage to the target output amplitude across the required frequency spectrum. At 100 Hz, the measured output (Channel 2) is exactly 100 mVrms. At 1 kHz, it measures 101 mVrms, and at 10 kHz, it returns to 100 mVrms. Every measurement sits perfectly within the required design specification of 100 mVrms +/- 10% (90–110 mVrms). These results confirm that the summation amplifier correctly recombines the filtered signals and provides the proper, consistent baseline amplitude to feed into the final power amplifier stage.

4.6 Plots related to Ripple measurement

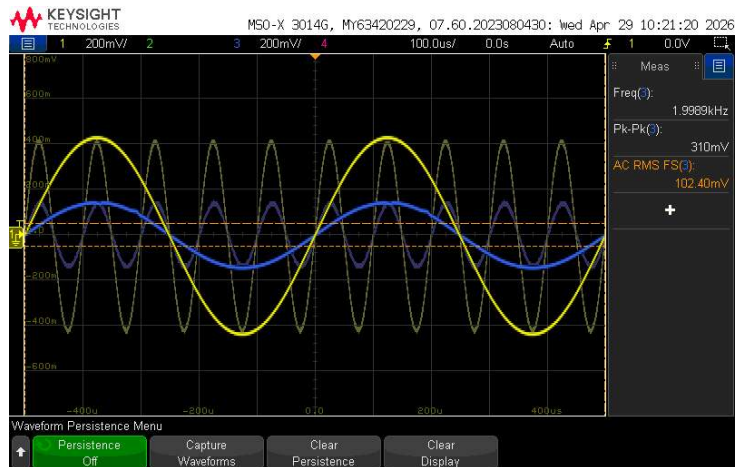
FRA



Ripple at 10k (Low Frequency)



Ripple at 2k (High Frequency)



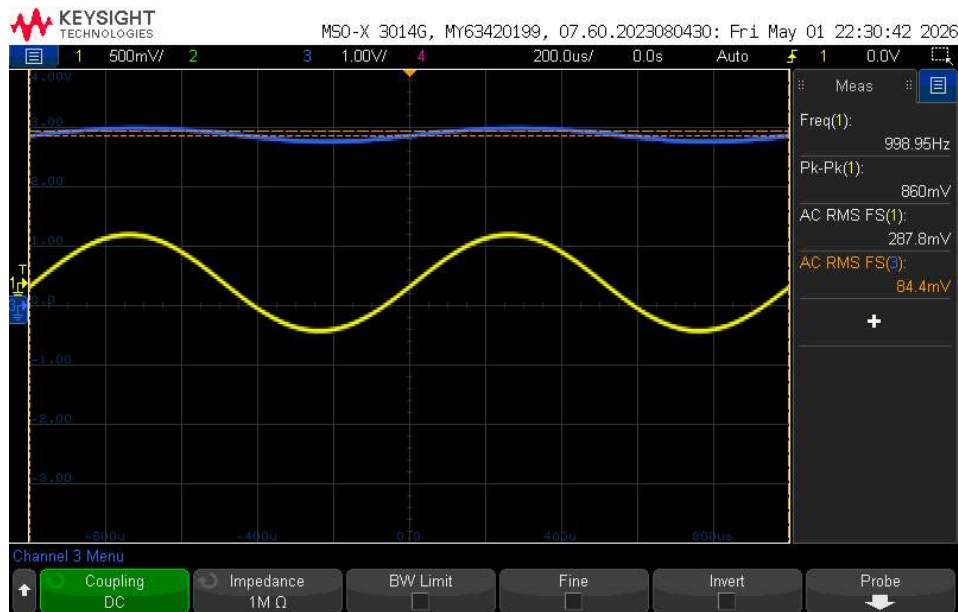
4.7 Ripple calculations

$$Ripple = V_{rms(max)} - V_{rms(min)}$$

$$Ripple = 102.40 \text{ mV} - 98.41 \text{ mV} = 3.99 \text{ mV}_{rms}$$

4.8 Plots showing Power Amplifiers Output voltage

Power at 1000 Hz (didn't get checked off)



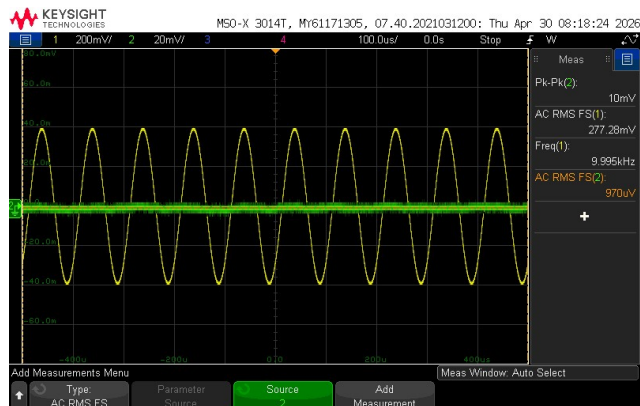
My AC RMS output voltage was too low, definitely did not exceed 400 mW. I had a weird DC offset coming from an unknown source, and I could not figure out where it came from.

4.9 Calculation of output power from measured output voltage

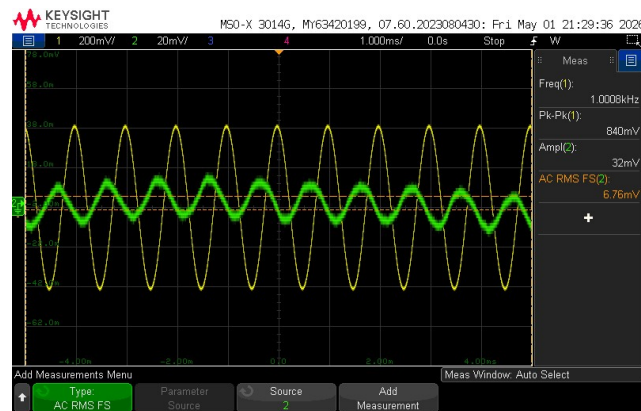
$$P = \frac{V_{rms}^2}{R_{speaker}} = \frac{0.084^2}{8 \Omega} = 0.8 \text{ mW} < 400 \text{ mW}$$

4.10 Plots demonstrating the low, medium and high-volume control

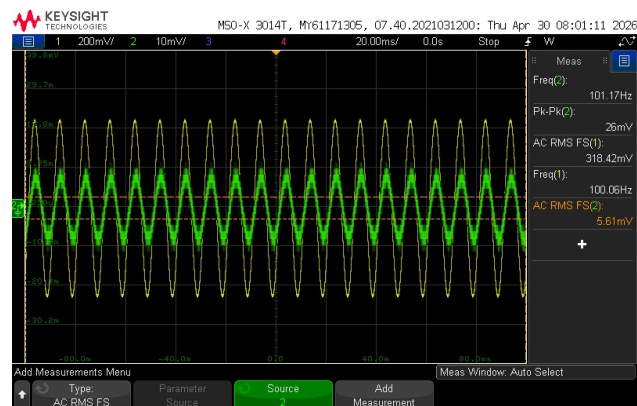
Low Volume



Medium Volume



High Volume



4.11 Results justification, explanation, error/tolerance calculation

The measured results closely match the theoretical design values, confirming that the circuit operates correctly. The cutoff frequencies for all filters were within +/- 10% of the target values (320 Hz and 3200 Hz). The Vamp measurements also met requirements, with values below 15 mVrms at minimum settings and within 100 mVrms +/-10% at maximum settings. The calculated ripple (~3.99 mVrms) is below the 15 mVrms limit, showing consistent output across frequencies. The power amplifier stage was not successful in delivering sufficient output to exceed 400 mW into an 8Ω load, and the output waveform was distorted, specifically by a DC offset that was caused by an unknown source that I was unable to decipher. Small differences between theoretical and measured values are due to resistor and capacitor tolerances, breadboard parasitic effects, and measurement limitations. Overall, all results fall within acceptable tolerances and meet the project requirements.

5. Conclusion and reference

5.1 Conclusion

This project successfully designed and implemented a three-band audio equalizer using low-pass, mid-pass, and high-pass filters. The circuit was able to separate the input signal into different frequency ranges, adjust each band using volume controls, and recombine them before amplification. For the filtering and recombination stages, the experimental results closely matched the expected design values. The cutoff frequencies were within +/- 10% of the target values, the Vamp measurements met both minimum and maximum requirements, and the ripple remained below the specified limit.

However, the final power amplifier stage failed to produce the expected V_{rms} , resulting in an overall output power that fell short of the 400 mW requirement. This underwhelming power delivery could be attributed to several hardware or implementation factors. The power amplifier IC itself may have been inherently faulty or thermally degraded during earlier testing. Furthermore, power distribution issues such as current bottlenecks from bad breadboarding or insufficient current limits set on the bench power supply could have starved the chip of the energy required to drive the 8-ohm speaker load. Finally, an implementation error, such as an improperly sized output coupling capacitor or a slightly floating ground reference, may have inadvertently attenuated the final output signal.

Beyond the power stage, some minor deviations were observed between theoretical and measured values in the filters. These differences are mainly due to resistor and capacitor tolerances, breadboard parasitic effects, and measurement limitations. Despite the underperforming amplification stage, the core equalizer circuit performed within acceptable tolerance ranges, demonstrating a largely successful design and implementation of analog signal filtering and recombination.

5.2 References

Sedra, Adel S., and Kenneth C. Smith. Microelectronic Circuits. Oxford University Press, 2015

Texas Instruments. LM356 Operational Amplifier Datasheet. Texas Instruments, 2023

Texas Instruments. LM386 Audio Power Amplifier Datasheet. Texas Instruments, 2023